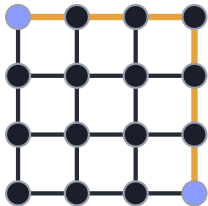
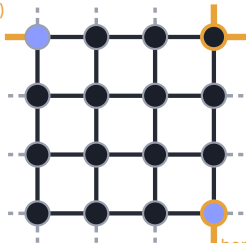


dashed stubs = wraparound links



4x4 mesh: 6 hops

hop 1
(wrap $\leftrightarrow$)



hop 2 (wrap $\updownarrow$)

4x4 torus: 2 hops